

Thermo-S1 · Part 1

Answer keys and marking schemes

Progress Tests 1.A → 3.C
Semester 1 · Academic Year 2026–2027

How to mark yourself

- Model answers are in **blue**, inside the same frames as on the paper. Your wording may differ; what matters is the content listed in the *marking scheme* below each exercise.
- Each circled item is worth the points shown. A numerical result without its unit, or with a wrong unit symbol, does not earn its item.
- Tick-box questions: 1 mark each. The “Why” line explains the answer and the trap behind the most tempting wrong option.
- Red boxes (*Examiner’s tip*) flag common traps, including the ones that cost marks in Test 1.
- Several boxes of the handout are blank (for example Boxes 4–6 and 8–18). The keys give the standard definitions; accept the wording used in class.

Lessons from Test 1, applied to thermodynamics

Test 1 (electricity)	In thermodynamics
“KV”: the marker wrote “big K stands for kelvins”.	Temperatures are in kelvins: “300 K”, never “300 °K”. The kilo prefix stays lowercase: kPa.
The unit of current is the ampere (A), an SI base unit, not “C s ^{−1} ”.	The kelvin (K) and the mole (mol) are SI base units. The pascal is derived: Pa = kg m ^{−1} s ^{−2} , and R is in J K ^{−1} mol ^{−1} .
Read the circuit diagram exactly as drawn (nodes, branches).	Read state diagrams exactly: the fixed variable (isotherm, isobar, isochore) decides which partial derivative a slope is.
Isolate the unknown first, carry units, handle powers of ten separately ($v = i/neS$, then 3.1×10^{-5} m/s).	Write $p = nRT/V$ first, then substitute SI values: 1 L = 10 ^{−3} m ³ , 1 bar = 10 ⁵ Pa, 1 cm ² = 10 ^{−4} m ² .

Numbers without a calculator. The papers give $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$ and $RT \approx 2.5 \times 10^3 \text{ J mol}^{-1}$ at 300 K. With the exact $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, the answers change by less than 1 %.

Progress Test 1.A — Thermo-S1

1.A · System and state variables

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

Convert the molar volume $V_m = 24.5 \text{ L mol}^{-1}$ into $\text{m}^3 \text{ mol}^{-1}$. Give the exact result, without rounding.

$1 \text{ L} = 1 \text{ dm}^3 = (10^{-1} \text{ m})^3 = 10^{-3} \text{ m}^3$	$V_m = 24.5 \times 10^{-3} \text{ m}^3 \text{ mol}^{-1}$
$V_m = 2.45 \times 10^{-2} \text{ m}^3 \text{ mol}^{-1}$	

Marking

①
0 1
☐ ☐

Marking scheme

① **1 pt** $2.45 \times 10^{-2} \text{ m}^3 \text{ mol}^{-1}$, with the unit written correctly.

Examiner's tip

A litre is a cubic decimetre: cube the conversion factor, $(10^{-1})^3 = 10^{-3}$. Every SI formula of this module ($pV = nRT$, ...) needs volumes in m^3 .

Exercise 2. Closed and isolated systems — definitions

(5 marks)

Define a closed system and an isolated system. For a gas enclosed in a cylinder, say what the walls must be like to prevent each kind of exchange with the surroundings. Is every closed system isolated?

Closed system: it exchanges no <i>matter</i> with its surroundings (n is constant). Isolated system: it exchanges neither <i>matter</i> nor <i>energy</i> with its surroundings. Walls needed for the gas to be isolated: • sealed walls, so that no matter is exchanged; • rigid walls and a fixed piston, so that no work is exchanged (pressure forces cannot move them); • thermally insulating walls, so that no heat is exchanged. An isolated system is necessarily closed, but the converse is false. For example, a sealed weather balloon is closed, yet it exchanges heat and work with the air around it.

Marking

①
0 1 2
☐ ☐ ☐

②
0 1 2
☐ ☐ ☐

③
0 1
☐ ☐

Marking scheme

- ① **2 pts** Closed: no exchange of matter (1 pt). Isolated: no matter and no energy (1 pt).
 ② **2 pts** Sealed, rigid or fixed, and insulating walls (1 pt if only two of the three).
 ③ **1 pt** Isolated $\Rightarrow$ closed, but closed $\nRightarrow$ isolated, with an example.

Exercise 3. Classifying quantities and systems

(4 marks)

Which of these quantities is intensive?

- ☐ the mass m ☐ the volume V
☒ the molar volume $V_m = V/n$ ☐ the energy E

Why: the quotient of two extensive quantities is intensive: doubling the system doubles both V and n , so V/n does not change.

A hot-air balloon, with its opening at the bottom, is:

- ☐ closed ☒ open ☐ isolated ☐ closed and isolated

Why: air enters and leaves through the opening, so the system exchanges matter (Fig. 1 of the notes).

The SI dimension of pressure is:

- ☒ $\text{ML}^{-1} \text{T}^{-2}$ ☐ MLT^{-2} ☐ $\text{ML}^2 \text{T}^{-2}$ ☐ $\text{ML}^{-2} \text{T}^{-2}$

Why: pressure is a force per unit area: $\text{MLT}^{-2}/\text{L}^2 = \text{ML}^{-1} \text{T}^{-2}$ (Tab. 1 of the notes).

 R is an intensive constant. Which relation is homogeneous in the extensive / intensive sense?

- ☐ $\frac{p}{T} = \frac{R}{V}$ ☐ $pV = RT$ ☒ $\frac{p}{T} = \frac{nR}{V}$ ☐ $p = nRT$

Why: p/T is intensive. So is nR/V (an extensive quantity divided by an extensive one). In the other options, one side doubles with the system and the other does not (Box 7).

Exercise 4. Composition of a solution

(4 marks)

A mass m of sodium chloride (molar mass M) is dissolved in water to obtain a volume V of solution. Compute the amount of substance n of solute, the mass concentration c_m and the molar concentration c . Check that $c_m = cM$.

$m = 5.85 \text{ g}$	$M = 58.5 \text{ g mol}^{-1}$	$V = 0.500 \text{ L}$
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$n = \frac{m}{M} = \frac{5.85 \text{ g}}{58.5 \text{ g mol}^{-1}}$	$n = 0.100 \text{ mol}$
$c_m = \frac{m}{V} = \frac{5.85 \text{ g}}{0.500 \text{ L}}$	$c_m = 11.7 \text{ g L}^{-1}$
$c = \frac{n}{V} = \frac{0.100 \text{ mol}}{0.500 \text{ L}}$	$c = 0.200 \text{ mol L}^{-1}$
Check: $cM = 0.200 \times 58.5 = 11.7 \text{ g L}^{-1} = c_m$. ✓ Both concentrations are intensive (quotients of extensive quantities).	

Marking

①	
0	1
②	
0	1
③	
0	1
④	
0	1

Marking scheme

- ① 1 pt $n = m/M = 0.100 \text{ mol}$.
- ② 1 pt $c_m = 11.7 \text{ g L}^{-1}$.
- ③ 1 pt $c = 0.200 \text{ mol L}^{-1}$.
- ④ 1 pt $c_m = cM$ checked.

Examiner's tip

Test 1, Ex. 4: write $n = m/M$ with letters first, then substitute *with units*. The units then cancel correctly by themselves: $\text{g}/(\text{g mol}^{-1}) = \text{mol}$.

Your score: / 14 $\rightarrow$ mark out of 20 = score $\times 20/14$ = / 20

Progress Test 1.B — Thermo-S1

1.B · Postulates of thermodynamics

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Temperature conversion

(1 mark)

Liquid nitrogen boils at $T = 77\text{ K}$ under atmospheric pressure. Express this temperature in degrees Celsius, exactly.

$\theta = T - 273.15 = 77 - 273.15$	$\theta = -196.15\text{ }^{\circ}\text{C}$
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Marking

①
0 1

Marking scheme

① 1 pt $-196.15\text{ }^{\circ}\text{C}$ (the minus sign is part of the answer).

Examiner's tip

The kelvin is written with a capital K and no degree sign: “77 K”, not “77 °K”. Test 1's marker noted: “big K stands for kelvins”.

Exercise 2. Thermodynamic equilibrium — postulates

(5 marks)

A system is in contact with surroundings at pressure p_0 and temperature T_0 . Give the two conditions of thermodynamic equilibrium, with the corresponding equalities. State the postulates of classical thermodynamics. When are the state variables of a system defined?

Thermal equilibrium: $T_{\text{system}} = T_0$. Mechanical equilibrium: $p_{\text{system}} = p_0$ (for a movable wall). Postulates. (1) A system left to itself (isolated, or in constant surroundings) evolves spontaneously towards an equilibrium state, and then no longer leaves it. (2) At equilibrium, the state of the system is completely described by a <i>small number</i> of macroscopic state variables (p, V, T, n). There is no need for the positions and velocities of all the particles. The state variables are defined (uniform in space and constant in time) only at equilibrium. Between two equilibrium states they may not be defined.
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Marking

①
0 1 2

②
0 1 2

③
0 1

Marking scheme

- ① **2 pts** $T = T_0$ (thermal) and $p = p_0$ (mechanical), 1 pt each.
 ② **2 pts** Evolution towards equilibrium (1 pt); a few state variables suffice (1 pt).
 ③ **1 pt** State variables are defined only at equilibrium.

Note

Box 8 is blank in the handout. These are the standard statements; accept the wording given in class.

Exercise 3. Equilibrium or not?

(4 marks)

A syringe with a free, sealed piston is left for a long time in a room at (p_0, T_0) . The enclosed gas then has:

- ☒ $p = p_0$ and $T = T_0$ ☐ $p > p_0$ and $T = T_0$
☐ $p = p_0$ and $T > T_0$ ☐ p and T not defined

Why: once equilibrium is reached, the state variables of the system equal those of the surroundings (Fig. 3 and remark (i)).

Just after a sudden push on the piston, the pressure of the gas:

- ☐ already has its final value ☒ is not defined: it is not uniform
☐ is equal to p_0 ☐ is zero

Why: it takes some time for p and T to become uniform and steady again: state variables exist only at equilibrium.

A metal rod has one end in boiling water and the other in melting ice. Its temperature profile no longer changes. The rod is:

- ☐ at thermodynamic equilibrium ☒ in a steady state, not at equilibrium
☐ isolated ☐ at equilibrium: nothing changes

Why: equilibrium requires both a steady state and homogeneity. Here T varies along the rod.

To describe a closed system at equilibrium, classical thermodynamics uses:

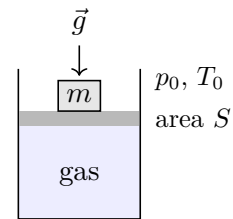
- ☐ all positions and velocities ☒ a few state variables (p, V, T)
☐ only its mass ☐ its whole past history

Why: this is the second postulate (remark (ii) of the notes).

Exercise 4. A loaded piston

(4 marks)

A vertical cylinder is closed by a frictionless piston of area S and negligible mass. The surroundings are at p_0 and T_0 . A mass m is placed on the piston. Find the pressure and the temperature of the gas once equilibrium is reached.



$$p_0 = 1.0 \times 10^5 \text{ Pa} \quad T_0 = 293 \text{ K} \quad m = 1.0 \text{ kg}$$

$$S = 2.0 \text{ cm}^2 \quad g = 10 \text{ m s}^{-2}$$

Equilibrium of the piston: $pS = p_0S + mg \Rightarrow p = p_0 + \frac{mg}{S}$

$$S = 2.0 \text{ cm}^2 = 2.0 \times (10^{-2} \text{ m})^2 = 2.0 \times 10^{-4} \text{ m}^2$$

$$\frac{mg}{S} = \frac{1.0 \times 10}{2.0 \times 10^{-4}} \text{ Pa} = 5.0 \times 10^4 \text{ Pa}$$

$$p = 1.5 \times 10^5 \text{ Pa} \quad \text{and, once thermal equilibrium is reached,}$$

$$T = T_0 = 293 \text{ K}$$

Marking

①		
0	1	
②		
0	1	
③		
0	1	
④		
0	1	

Marking scheme

- ① 1 pt Force balance on the piston: $p = p_0 + mg/S$ (literal).
- ② 1 pt $S = 2.0 \times 10^{-4} \text{ m}^2$.
- ③ 1 pt $p = 1.5 \times 10^5 \text{ Pa}$.
- ④ 1 pt $T = T_0 = 293 \text{ K}$ (thermal equilibrium).

Examiner's tip

$1 \text{ cm}^2 = (10^{-2} \text{ m})^2 = 10^{-4} \text{ m}^2$, not 10^{-2} . As in Test 1: keep letters until the end, then substitute with units.

Your score: /14 $\rightarrow$ mark out of 20 = score $\times$ 20/14 = /20

Progress Test 1.C — Thermo-S1

1.C · Equations of state and state functions

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

Convert the pressure $p = 1013 \text{ hPa}$ into bar. Give the exact result.

$p = 1013 \times 10^2 \text{ Pa} = 1.013 \times 10^5 \text{ Pa}$	and	$1 \text{ bar} = 10^5 \text{ Pa}$
$p = 1.013 \text{ bar}$		

Marking

①
0 1

Marking scheme

① 1 pt 1.013 bar.

Exercise 2. Equation of state and state function — definitions

(5 marks)

Define an equation of state. Write the total differential dV of the state function $V(T, p)$. Give the meaning of the partial derivative $\left(\frac{\partial V}{\partial p}\right)_T$, and its geometric interpretation in a state diagram.

An equation of state is a relation $f(p, V, T) = 0$ (at given n) between the state variables of a system at equilibrium. It characterises the substance and holds only at equilibrium.
$dV = \left(\frac{\partial V}{\partial T}\right)_p dT + \left(\frac{\partial V}{\partial p}\right)_T dp$
$\left(\frac{\partial V}{\partial p}\right)_T$ is the rate of change of V with p at fixed T. It is the slope of the isotherm $V(p)$ in the (p, V) diagram, and it is negative for a gas.

Marking

①
0 1 2
②
0 1 2
③
0 1

Marking scheme

- ① 2 pts Relation between p , V , T (1 pt), valid only at equilibrium and specific to the substance (1 pt).
- ② 2 pts Total differential with both partial derivatives and their fixed variables.
- ③ 1 pt Variation with p at fixed T ; slope of the isotherm.

Exercise 3. State diagrams

(4 marks)

The curve $V(p)$ measured at constant temperature is called:

- ☒ an isotherm ☐ an isobar ☐ an isochore ☐ an adiabat

Why: “iso-therm”: same temperature. At constant p it would be an isobar; at constant V , an isochore.

For a low-density gas, $\left(\frac{\partial V}{\partial p}\right)_T$ is:

- ☐ positive ☒ negative ☐ zero ☐ infinite

Why: at fixed T , compressing the gas (larger p) reduces its volume (Fig. 7 of the notes).

On the curve $V(T)$ at constant p , $\left(\frac{\partial V}{\partial T}\right)_p$ is:

- ☒ the slope of the curve ☐ the area under the curve
☐ the intercept at $T = 0$ ☐ the value of V

Why: a partial derivative is read as the slope of the state diagram in which the other variable is held fixed.

An equation of state is valid:

- ☐ during any transformation ☒ only at thermodynamic equilibrium
☐ only for ideal gases ☐ only at 0°C

Why: “By construction, the equation of state is valid only at thermodynamic equilibrium” (Box 9).

Exercise 4. Partial derivatives of a state function

(4 marks)

A gas obeys the equation of state $V = nRT/p$, which is given here. Compute $\left(\frac{\partial V}{\partial T}\right)_p$ and $\left(\frac{\partial V}{\partial p}\right)_T$ in the state below. Then compute the variation dV when T increases by dT and p increases by dp at the same time. Comment.

$n = 1.0 \text{ mol}$	$R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$	$T = 300 \text{ K}$	$p = 1.0 \times 10^5 \text{ Pa}$
	$dT = 3.0 \text{ K}$	$dp = 1.0 \times 10^3 \text{ Pa}$	

$$\left(\frac{\partial V}{\partial T}\right)_p = \frac{nR}{p} = \frac{1.0 \times 8.3}{1.0 \times 10^5} = 8.3 \times 10^{-5} \text{ m}^3 \text{ K}^{-1}$$

$$\left(\frac{\partial V}{\partial p}\right)_T = -\frac{nRT}{p^2} = -\frac{8.3 \times 300}{10^{10}} \approx -2.5 \times 10^{-7} \text{ m}^3 \text{ Pa}^{-1}$$

$$dV = 8.3 \times 10^{-5} \times 3.0 - 2.49 \times 10^{-7} \times 1.0 \times 10^3 = 2.49 \times 10^{-4} - 2.49 \times 10^{-4}$$

$$dV = 0$$

Dividing by V : $\frac{dV}{V} = \frac{dT}{T} - \frac{dp}{p} = 1\% - 1\% = 0$. T and p rise by the same relative amount, so their two effects on V cancel.

Marking

①		
0	1	
②		
0	1	
③		
0	1	
④		
0	1	

Marking scheme

- ① 1 pt $\left(\frac{\partial V}{\partial T}\right)_p = nR/p$ (literal).
- ② 1 pt $\left(\frac{\partial V}{\partial p}\right)_T = -nRT/p^2$ (literal).
- ③ 1 pt Both values, with units $\text{m}^3 \text{ K}^{-1}$ and $\text{m}^3 \text{ Pa}^{-1}$.
- ④ 1 pt $dV = 0$, explained.

Examiner's tip

The unit of a partial derivative is (unit of the numerator)/(unit of the variable), e.g. $\text{m}^3 \text{ K}^{-1}$. Writing it is exactly the “where are the units?” remark of Test 1.

Your score: /14 $\longrightarrow$ mark out of 20 = score $\times$ 20/14 = /20

Progress Test 2.A — Thermo-S1

2.A · Thermoelastic behaviour of condensed phases

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

The isothermal compressibility of water is $\chi_T = 4.5 \times 10^{-10} \text{ Pa}^{-1}$. Express it in bar^{-1} .

$1 \text{ bar} = 10^5 \text{ Pa, so } 1 \text{ Pa}^{-1} = 10^5 \text{ bar}^{-1}$	$\chi_T = 4.5 \times 10^{-10} \times 10^5 \text{ bar}^{-1}$
$\chi_T = 4.5 \times 10^{-5} \text{ bar}^{-1}$	

Marking

①
0 1

Marking scheme

① 1 pt $4.5 \times 10^{-5} \text{ bar}^{-1}$.

Examiner's tip

“Per pascal” to “per bar”: a bar is 10^5 times larger than a pascal, so the coefficient per bar is 10^5 times larger. A quick sanity check: water loses 4.5×10^{-5} of its volume per bar.

Exercise 2. Thermoelastic coefficients — definitions

(5 marks)

Define the isothermal compressibility χ_T and the isobaric expansion coefficient α of a condensed phase. Give their SI units. Explain the minus sign in χ_T , and why both definitions divide by V .

$\chi_T = -\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T$ in Pa^{-1}	$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p$ in K^{-1}
Minus sign: when p increases at fixed T, V decreases, so $\left(\frac{\partial V}{\partial p} \right)_T < 0$. The minus sign makes $\chi_T > 0$.	
Division by V: $\left(\frac{\partial V}{\partial p} \right)_T$ and $\left(\frac{\partial V}{\partial T} \right)_p$ are extensive. Dividing by V gives intensive coefficients, which do not depend on the size of the sample.	

Marking

①
0 1 2

②
0 1 2

③
0 1

Marking scheme

- ① **2 pts** $\chi_T = -\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T$, with the fixed variable T written.
② **2 pts** $\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p$, with the fixed variable p written.
③ **1 pt** Units Pa^{-1} and K^{-1} ; sign and intensive character explained.

Exercise 3. Condensed phases

(3 marks)

For a liquid or a solid, the isothermal compressibility χ_T is:

- | | |
|---|-------------------------------------|
| ☒ very small ($\chi_T \approx 0$) | ☐ very large |
| ☐ negative | ☐ infinite |

Why: “condensed phases are weakly compressible”: their volume hardly changes when p increases.

Expressed with the mass density ρ , χ_T equals:

- | | |
|--|--|
| ☒ $\frac{1}{\rho} \left(\frac{\partial \rho}{\partial p} \right)_T$ | ☐ $-\frac{1}{\rho} \left(\frac{\partial \rho}{\partial p} \right)_T$ |
| ☐ $\rho \left(\frac{\partial \rho}{\partial p} \right)_T$ | ☐ $-\frac{1}{V} \left(\frac{\partial \rho}{\partial p} \right)_T$ |

Why: for a closed system $\rho = m/V$, so compressing increases ρ . The minus sign disappears (equation (7) of the notes).

Over a restricted temperature range, the coefficient α of a condensed phase can be considered:

- | | |
|--|-----------------------------------|
| ☒ constant | ☐ zero |
| ☐ proportional to p | ☐ negative |

Why: remark (ii) of Box 11. This is what allows the double integration of Box 13.

Exercise 4. Expansion and compression of water

(5 marks)

The coefficients of water are assumed constant. (a) A volume V_0 of water is heated at constant pressure from θ_1 to θ_2 . Estimate the change of volume ΔV . (b) Water is compressed at constant temperature from p_1 to p_2 . Estimate its relative change of volume $\Delta V/V$.

$$\alpha = 2.0 \times 10^{-4} \text{ K}^{-1} \quad \chi_T = 5.0 \times 10^{-10} \text{ Pa}^{-1} \quad V_0 = 250 \text{ mL}$$

$$\theta_1 = 20^\circ\text{C} \quad \theta_2 = 70^\circ\text{C} \quad p_1 = 1 \text{ bar} \quad p_2 = 101 \text{ bar}$$

(a) At fixed p , $\frac{dV}{V} = \alpha dT$. For a small change, $\Delta V \approx \alpha V_0 \Delta T$ with $\Delta T = 70 - 20 = 50 \text{ K}$ (a difference in $^\circ\text{C}$ equals a difference in K):

$$\Delta V \approx 2.0 \times 10^{-4} \times 250 \text{ mL} \times 50 \quad \Delta V \approx 2.5 \text{ mL} \quad (1\% \text{ of } V_0)$$

(b) At fixed T , $\frac{dV}{V} = -\chi_T dp$, so $\frac{\Delta V}{V} \approx -\chi_T \Delta p$ with

$$\Delta p = 100 \text{ bar} = 1.0 \times 10^7 \text{ Pa:}$$

$$\frac{\Delta V}{V} \approx -5.0 \times 10^{-10} \times 1.0 \times 10^7 \quad \frac{\Delta V}{V} \approx -5.0 \times 10^{-3} = -0.5\%$$

Even 100 bar changes the volume by only 0.5%: water is almost incompressible.

Marking

①	0	1
②	0	1
③	0	1
④	0	1
⑤	0	1

Marking scheme

- ① 1 pt $\Delta V \approx \alpha V_0 \Delta T$ (literal).
- ② 1 pt $\Delta T = 50 \text{ K}$ and $\Delta V \approx 2.5 \text{ mL}$.
- ③ 1 pt $\Delta V/V \approx -\chi_T \Delta p$ (literal, with the minus sign).
- ④ 1 pt $\Delta p = 1.0 \times 10^7 \text{ Pa}$.
- ⑤ 1 pt $\Delta V/V \approx -5.0 \times 10^{-3}$, interpreted.

Examiner's tip

A *difference* of temperature has the same value in K and in $^\circ\text{C}$, so here $\Delta T = 50 \text{ K}$. Only absolute temperatures (as in $pV = nRT$) must be converted.

Your score: / 14 $\longrightarrow$ mark out of 20 = score $\times$ 20/14 = / 20

Progress Test 2.B — Thermo-S1

2.B · Thermoelastic behaviour of gases

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

A gas has a piezothermic coefficient $\beta = 3.66 \times 10^{-3} \text{ K}^{-1}$. Express β in $^{\circ}\text{C}^{-1}$.

A temperature *difference* of 1 K equals a difference of 1 $^{\circ}\text{C}$, so
 $\beta = 3.66 \times 10^{-3} \text{ }^{\circ}\text{C}^{-1}$ (same value).

Marking

①
0 1

Marking scheme

① 1 pt $3.66 \times 10^{-3} \text{ }^{\circ}\text{C}^{-1}$ (no +273.15 here).

Examiner's tip

A coefficient “per kelvin” is defined through a derivative, i.e. a temperature *difference*. Adding 273.15 would be a classic mistake.

Exercise 2. Piezothermic coefficient — definition

(5 marks)

Define the piezothermic coefficient β of a gas. Give its formula, the physical variation it measures, its sign and its SI unit. How does it differ from α ?

$\beta = \frac{1}{p} \left(\frac{\partial p}{\partial T} \right)_V$

It measures the **relative increase of the pressure per kelvin at constant volume**, for example a gas heated in a rigid, sealed container. Heating increases the pressure, so $\beta > 0$.

Unit: K^{-1} . By contrast, $\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p$ describes the change of *volume* at constant *pressure*.

Marking

①
0 1 2
②
0 1 2
③
0 1

Marking scheme

- ① **2 pts** $\beta = \frac{1}{p} \left(\frac{\partial p}{\partial T} \right)_V$, with V fixed.
② **2 pts** Meaning: change of p with T at constant V (1 pt); $\beta > 0$ (1 pt).
③ **1 pt** K^{-1} and the difference with α .

Exercise 3. Gases at constant volume

(3 marks)

A gas enclosed in a rigid, sealed container is heated. Its pressure:

☒ increases ☐ decreases ☐ does not change ☐ becomes zero

Why: $\left(\frac{\partial p}{\partial T} \right)_V > 0$, i.e. $\beta > 0$ (equation (10) of the notes).

The coefficient β measures the variation of:

☐ V at fixed p ☒ p at fixed V
☐ T at fixed V ☐ V at fixed T

Why: see Tab. 3 of the notes. α : V at fixed p ; χ_T : V at fixed T .

For a car tyre (airtight, rigid rim), the variables held fixed are:

☒ n and V ☐ n and p ☐ T and V ☐ p and V

Why: Tab. 5: the tyre is airtight (n fixed) and the rim is rigid (V fixed). T and p rise because of friction.

Exercise 4. Tyre pressure

(5 marks)

A car tyre, whose volume is fixed, is inflated to p_1 at T_1 . While driving, its temperature rises by ΔT . Using the value of β given below (assumed constant), estimate the pressure increase Δp and the final pressure p_2 .

$p_1 = 2.0 \text{ bar}$	$T_1 = 300 \text{ K}$	$\Delta T = 30 \text{ K}$	$\beta = \frac{1}{300} \text{ K}^{-1}$
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At fixed V : $dp = \beta p dT$, so for a small change $\Delta p \approx \beta p_1 \Delta T$.

$p_1 = 2.0 \text{ bar} = 2.0 \times 10^5 \text{ Pa}$ $\Delta p \approx \frac{1}{300} \times 2.0 \times 10^5 \times 30$

$\Delta p \approx 2.0 \times 10^4 \text{ Pa} = 0.20 \text{ bar}$

$p_2 \approx 2.2 \text{ bar}$: a rise of 10 %, the same as $\Delta T/T_1$. (For such a gas $\beta = 1/T$, see 3.A, and p/T stays constant: $p_2 = p_1 T_2/T_1$ exactly.)

Marking

①
0 1
☐ ☐
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☐ ☐
④
0 1
☐ ☐
⑤
0 1
☐ ☐

Marking scheme

- ① 1 pt $\Delta p \approx \beta p \Delta T$ from the definition at constant V .
- ② 1 pt $p_1 = 2.0 \times 10^5 \text{ Pa}$.
- ③ 1 pt $\Delta p \approx 2.0 \times 10^4 \text{ Pa}$.
- ④ 1 pt $\Delta p = 0.20 \text{ bar}$ and $p_2 \approx 2.2 \text{ bar}$.
- ⑤ 1 pt Relative increase 10 % = $\Delta T/T_1$ noted.

Examiner's tip

Tyre pressures must be checked *cold*: a hot tyre reads about 10 % higher.

Your score: / 14 → mark out of 20 = score $\times$ 20/14 = / 20

Progress Test 2.C — Thermo-S1

2.C · Thermoelastic coefficients: principle and method

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Quick calculation

(1 mark)

At $p = 1.0 \times 10^5$ Pa, a gas has $\alpha = 3.3 \times 10^{-3} \text{ K}^{-1}$ and $\chi_T = 1.0 \times 10^{-5} \text{ Pa}^{-1}$. Give its piezothermic coefficient β .

$$\beta = \frac{\alpha}{p\chi_T} = \frac{3.3 \times 10^{-3}}{1.0 \times 10^5 \times 1.0 \times 10^{-5}} \quad \boxed{\beta = 3.3 \times 10^{-3} \text{ K}^{-1}}$$

Marking

①
0 1
☐ ☐

Marking scheme

① 1 pt $\beta = 3.3 \times 10^{-3} \text{ K}^{-1}$ (the product $p\chi_T$ is dimensionless).

Exercise 2. Relation between the coefficients

(5 marks)

State the cyclic relation between the partial derivatives of p , V and T . Use it to show that $\beta = \frac{\alpha}{p\chi_T}$.

For the equation of state $f(p, V, T) = 0$:

$$\left(\frac{\partial p}{\partial T}\right)_V \left(\frac{\partial T}{\partial V}\right)_p \left(\frac{\partial V}{\partial p}\right)_T = -1$$

With $\left(\frac{\partial T}{\partial V}\right)_p = 1/\left(\frac{\partial V}{\partial T}\right)_p$:

$$\left(\frac{\partial p}{\partial T}\right)_V = -\frac{(\partial V/\partial T)_p}{(\partial V/\partial p)_T} = -\frac{\alpha V}{-\chi_T V} = \frac{\alpha}{\chi_T}$$

Hence $\beta = \frac{1}{p} \left(\frac{\partial p}{\partial T}\right)_V$ gives $\boxed{\beta = \frac{\alpha}{p\chi_T}}$

Marking

①
0 1 2
☐ ☐ ☐
②
0 1 2
☐ ☐ ☐
③
0 1
☐ ☐

Marking scheme

- ① 2 pts Cyclic relation, with the correct fixed variables and -1 on the right.
 ② 2 pts $(\partial p/\partial T)_V = \alpha/\chi_T$, derived.
 ③ 1 pt $\beta = \alpha/(p\chi_T)$.

Examiner's tip

The product of the three derivatives is -1 , not $+1$. Check the sign on an ideal gas: $\alpha = 1/T$ and $\chi_T = 1/p$ give $\beta = 1/T > 0$.

Exercise 3. Method in thermodynamics

(3 marks)

To obtain the equation of state from α and χ_T , one must:

- ☐ differentiate ☒ integrate ☐ multiply them ☐ average them

Why: knowing α and χ_T as functions of (T, p) , a double integration rebuilds $V(T, p)$ (Fig. 8 of the notes).

To obtain α from a known equation of state, one must:

- ☒ differentiate ☐ integrate ☐ invert it ☐ plot it

Why: express $V(T, p)$, then compute the partial derivative $(\partial V / \partial T)_p$.

The three coefficients α , χ_T and β are:

- ☐ independent of each other ☒ linked by $\beta = \alpha / (p\chi_T)$
☐ always equal ☐ all dimensionless

Why: only two of them are independent; the third follows from the cyclic relation.

Exercise 4. From an equation of state to the coefficients

(5 marks)

A gas obeys the equation of state $p(V - nb) = nRT$, where the co-volume b is a positive constant. Compute α , χ_T and β , then check that $\beta = \alpha / (p\chi_T)$. What do you find when $b \rightarrow 0$?

$V = \frac{nRT}{p} + nb:$	$\left(\frac{\partial V}{\partial T}\right)_p = \frac{nR}{p}$ so	$\alpha = \frac{nR}{pV}$	$\left(\frac{\partial V}{\partial p}\right)_T = -\frac{nRT}{p^2}$ so
$\chi_T = \frac{nRT}{p^2V}$			
$p = \frac{nRT}{V - nb}:$	$\left(\frac{\partial p}{\partial T}\right)_V = \frac{nR}{V - nb} = \frac{p}{T}$ so	$\beta = \frac{1}{T}$	
Check: $\frac{\alpha}{p\chi_T} = \frac{nR}{pV} \times \frac{p^2V}{pnRT} = \frac{1}{T} = \beta. \checkmark$			
When $b \rightarrow 0$: $V = nRT/p$, so $\alpha = \frac{1}{T}$ and $\chi_T = \frac{1}{p}$, the values for the ideal gas (3.A).			

Marking

①
0 1
②
0 1
③
0 1
④
0 1
⑤
0 1

Marking scheme

- ① 1 pt $\alpha = nR/(pV)$.
 ② 1 pt $\chi_T = nRT/(p^2V)$.
 ③ 1 pt $\beta = 1/T$.
 ④ 1 pt Relation checked.
 ⑤ 1 pt Limit $b \rightarrow 0$: $\alpha = 1/T$, $\chi_T = 1/p$.

Your score: / 14 $\rightarrow$ mark out of 20 = score $\times$ 20/14 = / 20

Progress Test 3.A — Thermo-S1

3.A · Low-density gases: the historical laws

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

The product pV of a gas equals 2.0 bar L. Express it in joules.

$pV = 2.0 \times 10^5 \text{ Pa} \times 1.0 \times 10^{-3} \text{ m}^3$ ($\text{Pa m}^3 = \text{N m} = \text{J}$)	$pV = 2.0 \times 10^2 \text{ J}$
--	----------------------------------

Marking

①
0 1

Marking scheme

① 1 pt 200 J, with the unit.

Examiner's tip

pV has the dimension of an energy, which is why R is in $\text{J K}^{-1} \text{mol}^{-1}$. Useful to remember:
1 bar L = 100 J.

Exercise 2. Boyle–Mariotte law

(5 marks)

State the Boyle–Mariotte law: the conditions under which it holds, and its formula. Deduce from it the isothermal compressibility χ_T of a low-density gas.

Boyle–Mariotte (1662): for a low-density gas at fixed temperature T (and fixed n), the product pV is constant, whatever the gas: $pV = \text{const.}$ or $p_1 V_1 = p_2 V_2$ ($T_1 = T_2$) So $V = C/p$ and $\left(\frac{\partial V}{\partial p}\right)_T = -\frac{C}{p^2} = -\frac{V}{p}$, hence $\chi_T = -\frac{1}{V} \left(\frac{\partial V}{\partial p}\right)_T = \frac{1}{p}$ in Pa^{-1}. The gas becomes less compressible as the pressure increases.	
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Marking

①
0 1 2
②
0 1 2
③
0 1

Marking scheme

- ① **2 pts** Statement: fixed T and n , low density (1 pt); $pV = \text{const.}$ (1 pt).
② **2 pts** $(\partial V/\partial p)_T = -V/p$, derived.
③ **1 pt** $\chi_T = 1/p$, in Pa^{-1} .

Exercise 3. The three historical laws

(3 marks)

Charles's law holds at fixed:

- ☐ n and T ☒ n and p ☐ n and V ☐ p and V

Why: Tab. 4: Charles (1787), fixed n and p , free V and T : $V/T = \text{const.}$

Gay-Lussac's law states that, at fixed n and V :

- ☐ $pV = \text{const.}$ ☐ $V/T = \text{const.}$ ☒ $p/T = \text{const.}$ ☐ $pT = \text{const.}$

Why: equation (14) of the notes: $p_1/T_1 = p_2/T_2$ if $V_1 = V_2$.

For a low-density gas, the isobaric expansion coefficient is:

- ☒ $\alpha = 1/T$ ☐ $\alpha = 1/p$ ☐ $\alpha = T$ ☐ $\alpha = 0$

Why: $V/T = \text{const.}$ gives $(\partial V/\partial T)_p = V/T$, so $\alpha = 1/T$ (Box 15).

Exercise 4. A rising bubble and a heated balloon

(5 marks)

(a) An air bubble of volume V_1 is released at a depth where the pressure is p_1 . It rises to the surface, where the pressure is p_2 , at constant temperature. Find its volume V_2 at the surface. (b) A balloon of volume V at θ_1 is heated to θ_2 at constant pressure. Find its new volume V' .

$$\begin{array}{lll} \text{(a)} & V_1 = 2.0 \text{ cm}^3 & p_1 = 3.0 \text{ bar} & p_2 = 1.0 \text{ bar} \\ & V = 3.0 \text{ L} & \theta_1 = 27^\circ\text{C} & \theta_2 = 77^\circ\text{C} \end{array}$$

(a) Fixed n and T , so Boyle–Mariotte applies: $p_1 V_1 = p_2 V_2$, i.e.

$$V_2 = \frac{p_1 V_1}{p_2} = \frac{3.0}{1.0} \times 2.0 \text{ cm}^3 \quad V_2 = 6.0 \text{ cm}^3$$

(b) Fixed n and p , so Charles applies: $\frac{V}{T_1} = \frac{V'}{T_2}$, i.e. $V' = V \frac{T_2}{T_1}$, with **absolute** temperatures $T_1 = 300 \text{ K}$ and $T_2 = 350 \text{ K}$:

$$V' = 3.0 \text{ L} \times \frac{350}{300} \quad V' = 3.5 \text{ L}$$

Marking

①	0	1
②	0	1
③	0	1
④	0	1
⑤	0	1

Marking scheme

- ① 1 pt $V_2 = p_1 V_1 / p_2$ (literal).
- ② 1 pt $V_2 = 6.0 \text{ cm}^3$.
- ③ 1 pt $V' = V T_2 / T_1$ (literal).
- ④ 1 pt Temperatures converted into kelvins: 300 K and 350 K.
- ⑤ 1 pt $V' = 3.5 \text{ L}$.

Examiner's tip

The laws of Charles and Gay-Lussac need absolute temperatures. Using $^\circ\text{C}$ would give $V' = 3.0 \times 77/27 \approx 8.6 \text{ L}$, which is absurd.

Your score: / 14 → mark out of 20 = score $\times$ 20/14 = / 20

Progress Test 3.B — Thermo-S1

3.B · The ideal gas and Clapeyron's law

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Units

(1 mark)

Express the unit of the gas constant R , $\text{J K}^{-1} \text{mol}^{-1}$, in SI base units.

$1 \text{ J} = 1 \text{ N m} = 1 \text{ kg m}^2 \text{ s}^{-2}$	$[R] = \text{kg m}^2 \text{ s}^{-2} \text{ K}^{-1} \text{mol}^{-1}$
---	---

Marking

①
0 1
☐ ☐

Marking scheme

① 1 pt $\text{kg m}^2 \text{ s}^{-2} \text{ K}^{-1} \text{mol}^{-1}$.

Exercise 2. The ideal gas — definition and Clapeyron's law

(5 marks)

Define the ideal gas model. Write Clapeyron's law, giving the SI unit of every quantity and the value of R .

The ideal gas is the limit model towards which <i>all</i> gases tend when their particle density n/V is low enough. Its particles have no interactions other than collisions with the walls of the container. Clapeyron (1834): $pV = nRT$ p in Pa, V in m^3, n in mol, T in K, $R = 8.314 \text{ J K}^{-1} \text{mol}^{-1}$.

Marking

①
0 1 2
☐ ☐ ☐

②
0 1 2
☐ ☐ ☐

③
0 1
☐ ☐

Marking scheme

- ① 2 pts Low-density limit (1 pt); no interactions between particles (1 pt).
- ② 2 pts $pV = nRT$.
- ③ 1 pt The five units, with T in kelvins, and $R \approx 8.31 \text{ SI}$.

Exercise 3. Controlled and free variables

(3 marks)

For a hot-air balloon (opening at the bottom, rigid envelope), the variables held fixed are:

- ☐ n and V ☒ p and V ☐ n and T ☐ n and p

Why: Tab. 5: $p = p_{\text{atm}}$ (mechanical equilibrium) and V is fixed by the rigid envelope. n and T are free: heating drives air out.

A weather balloon (sealed, flexible) rises; the outside pressure decreases while its temperature stays about the same. Its volume:

- ☒ increases ☐ decreases ☐ stays the same ☐ becomes zero

Why: n is fixed and $p = p_{\text{atm}}$ decreases. With $pV = nRT$ and T about constant, V increases.

1.0 mol of gas at 300 K and 1.0×10^6 Pa occupies 2.5 L. With $RT \approx 2.5 \times 10^3 \text{ J mol}^{-1}$, its compressibility factor $Z = pV/(nRT)$ is about:

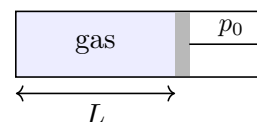
- ☐ 0.1 ☒ 1.0 ☐ 10 ☐ 2.5

Why: $Z = \frac{1.0 \times 10^6 \times 2.5 \times 10^{-3}}{2.5 \times 10^3} = 1.0$: the gas behaves as an ideal gas. (Compare the worked example of the notes: 1 mol of air in 2 L at 2 bar and 25 °C is far from ideal.)

Exercise 4. Piston and cylinder

(5 marks)

A horizontal cylinder of cross-section S contains n moles of air, treated as an ideal gas. The piston is initially at L_1 and the temperature is T_1 .



(1) Compute p_1 . (2) The gas is compressed isothermally to L_2 : compute p_2 . (3) The piston is locked at L_2 and the gas is heated to T_3 : compute p_3 . (4) The piston is released against the atmosphere p_0 , at $T_4 = T_3$: compute V_4 and L_4 .

$$S = 50 \text{ cm}^2 \quad n = 0.10 \text{ mol} \quad L_1 = 50 \text{ cm} \quad L_2 = 25 \text{ cm} \quad T_1 = 300 \text{ K}$$

$$T_3 = 450 \text{ K} \quad p_0 = 1.0 \times 10^5 \text{ Pa} \quad RT \approx 2.5 \times 10^3 \text{ J mol}^{-1} \text{ at } 300 \text{ K}$$

$$(1) V_1 = SL_1 = 50 \times 10^{-4} \times 0.50 = 2.5 \times 10^{-3} \text{ m}^3$$

$$p_1 = \frac{nRT_1}{V_1} = \frac{0.10 \times 2.5 \times 10^3}{2.5 \times 10^{-3}} \quad p_1 = 1.0 \times 10^5 \text{ Pa}$$

$$(2) n \text{ and } T \text{ fixed: } p_1 V_1 = p_2 V_2, \text{ so } p_2 = p_1 \frac{L_1}{L_2} \quad p_2 = 2.0 \times 10^5 \text{ Pa}$$

$$(3) n \text{ and } V \text{ fixed: } \frac{p}{T} = \text{const.}, \text{ so } p_3 = p_2 \frac{T_3}{T_1} = 2.0 \times 10^5 \times \frac{450}{300}$$

$$p_3 = 3.0 \times 10^5 \text{ Pa}$$

$$(4) n, T \text{ and } p = p_0 \text{ fixed: } V_4 = \frac{nRT_4}{p_0}, \text{ with}$$

$$RT_4 = 1.5 \times 2.5 \times 10^3 = 3.75 \times 10^3 \text{ J mol}^{-1}:$$

$$V_4 = \frac{0.10 \times 3.75 \times 10^3}{1.0 \times 10^5} \quad V_4 = 3.75 \times 10^{-3} \text{ m}^3$$

$$L_4 = \frac{V_4}{S} = \frac{3.75 \times 10^{-3}}{5.0 \times 10^{-3}} \quad L_4 = 0.75 \text{ m}$$

Marking

①	0	1
②	0	1
③	0	1
④	0	1
⑤	0	1

Marking scheme

- ① 1 pt $p_1 = 1.0 \times 10^5 \text{ Pa}$ (with S and L_1 converted to SI).
- ② 1 pt $p_2 = 2.0 \times 10^5 \text{ Pa}$.
- ③ 1 pt $p_3 = 3.0 \times 10^5 \text{ Pa}$.
- ④ 1 pt $V_4 = 3.75 \times 10^{-3} \text{ m}^3$ (3.75 L).
- ⑤ 1 pt $L_4 = 75 \text{ cm}$.

Examiner's tip

As in Tab. 5 of the notes, first list which variables are fixed at each step: then the right law (pV , p/T or V/T constant) follows at once.

Your score: / 14 → mark out of 20 = score $\times$ 20/14 = / 20

Progress Test 3.C — Thermo-S1

3.C · Equations of state of real gases

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

The co-volume of carbon dioxide is $b = 4.27 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1}$. Express it in $\text{cm}^3 \text{ mol}^{-1}$.

$1 \text{ m}^3 = (10^2 \text{ cm})^3 = 10^6 \text{ cm}^3$	$b = 4.27 \times 10^{-5} \times 10^6 \text{ cm}^3 \text{ mol}^{-1}$
$b = 42.7 \text{ cm}^3 \text{ mol}^{-1}$	

Marking

①
0 1

Marking scheme

① 1 pt $42.7 \text{ cm}^3 \text{ mol}^{-1}$.

Exercise 2. The Van der Waals model

(5 marks)

Write the Van der Waals equation of state for n moles. Explain the physical meaning of the term involving b and of the term involving a . What does the equation reduce to when $a \rightarrow 0$ and $b \rightarrow 0$?

$\left(p + a \frac{n^2}{V^2}\right) (V - nb) = nRT$ nb is the co-volume. Real molecules have a non-zero size, so the volume actually available to them is $V - nb$ (the excluded volume is about four times the molecules' own volume). This repulsive effect raises the pressure. $a n^2/V^2$ is the internal pressure. Molecules attract each other, so a molecule heading for the wall is held back by its neighbours and the measured pressure is lower. The term is proportional to $(n/V)^2$: the number of molecules reaching the wall and the number of neighbours pulling each one back both scale with n/V. When $a \rightarrow 0$ and $b \rightarrow 0$: $pV = nRT$, the ideal gas.

Marking

①
0 1 2
②
0 1 2
③
0 1

Marking scheme

- ① 2 pts The equation, with n^2/V^2 and $V - nb$.
 ② 2 pts b : excluded volume or finite size (1 pt); a : attraction, internal pressure (1 pt).
 ③ 1 pt Limit: ideal gas.

Exercise 3. The Amagat diagram

(4 marks)

In the Amagat diagram, when $p \rightarrow 0$ the compressibility factor Z tends to:

- ☒ 1 ☐ 0 ☐ $+\infty$ ☐ -1

Why: at very low density every gas behaves as an ideal gas, for which $Z = 1$.

At moderate pressure, $Z < 1$ for most gases because:

- ☒ attractive interactions dominate ☐ the molecules' size dominates
☐ the gas is ideal ☐ the temperature is too high

Why: attractions reduce the pressure (or the volume) below the ideal value. At high pressure the excluded volume wins and $Z > 1$.

The Boyle temperature T_B is the temperature at which:

- ☒ $Z \approx 1$ over a wide range of p ☐ $Z = 0$
☐ the gas liquefies ☐ $a = 0$

Why: at T_B the attractive and repulsive corrections compensate over a wide range of pressures (Fig. 9).

According to Tab. 7, the gas with the strongest intermolecular attractions, and the highest critical temperature, is:

- ☐ He ☐ H₂ ☐ N₂ ☒ H₂O

Why: $a(\text{H}_2\text{O}) = 0.554 \text{ Pa m}^6 \text{ mol}^{-2}$ and $T_c = 647 \text{ K}$: a and T_c are correlated.

Exercise 4. Carbon dioxide under pressure

(4 marks)

One mole of CO₂ occupies the volume V at temperature T . Compute (a) the pressure p_{id} predicted by the ideal gas law; (b) the pressure p predicted by the Van der Waals equation; (c) the corresponding compressibility factor Z . Interpret.

$$n = 1.0 \text{ mol} \quad V = 0.50 \text{ L} \quad T = 300 \text{ K} \quad RT \approx 2.5 \times 10^3 \text{ J mol}^{-1}$$

$$a = 0.36 \text{ Pa m}^6 \text{ mol}^{-2} \quad b = 4.0 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1} \quad (\text{rounded values for CO}_2)$$

(a) $p_{\text{id}} = \frac{nRT}{V} = \frac{2.5 \times 10^3}{5.0 \times 10^{-4}} \quad p_{\text{id}} = 5.0 \times 10^6 \text{ Pa} \quad (50 \text{ bar})$

(b) $p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$, with

$$V - nb = 5.0 \times 10^{-4} - 0.40 \times 10^{-4} = 4.6 \times 10^{-4} \text{ m}^3:$$

$$\frac{nRT}{V - nb} = \frac{2.5 \times 10^3}{4.6 \times 10^{-4}} \approx 5.4 \times 10^6 \text{ Pa}$$

$$a \frac{n^2}{V^2} = \frac{0.36}{(5.0 \times 10^{-4})^2} = 1.44 \times 10^6 \text{ Pa}$$

$$p \approx 4.0 \times 10^6 \text{ Pa} \quad (40 \text{ bar})$$

(c) $Z = \frac{pV}{nRT} = \frac{p}{p_{\text{id}}} \quad Z \approx 0.80 < 1$. Attraction dominates, as on the Amagat diagram at moderate pressure.

Marking

①		
0	1	
②		
0	1	
③		
0	1	
④		
0	1	

Marking scheme

- ① 1 pt $p_{\text{id}} = 5.0 \times 10^6 \text{ Pa}$.
- ② 1 pt Van der Waals equation solved for p (literal).
- ③ 1 pt $p \approx 4.0 \times 10^6 \text{ Pa}$.
- ④ 1 pt $Z \approx 0.8$, attractions dominate.

Examiner's tip

Convert first: $0.50 \text{ L} = 5.0 \times 10^{-4} \text{ m}^3$. Then $(5.0 \times 10^{-4})^2 = 25 \times 10^{-8} = 2.5 \times 10^{-7} \text{ m}^6$: square the mantissa *and* the power of ten.

Your score: / 14 → mark out of 20 = score $\times$ 20/14 = / 20